

Research Notes on Hidden Markov model

Hidden Markov model

$$P(O | \lambda) = \sum_Q P(O | Q, \lambda) P(Q | \lambda)$$

>> Section 1

In its discrete form, a hidden Markov process can be visualized as a generalization of the urn problem with replacement (where each item from the urn is returned to the original urn before the next step).

>> Section 2

Bayesian modeling of the transitions probabilities Hidden Markov models are generative models, in which the joint distribution of observations and hidden states, or equivalently both the prior distribution of hidden states (the transition probabilities) and conditional distribution of observations given states (the emission probabilities), is modeled. The above algorithms implicitly assume a uniform prior distribution over the transition probabilities. However, it is also possible to create hidden Markov models with other types of prior distributions. An obvious candidate, given the categorical distribution of the transition probabilities, is the Dirichlet distribution, which is the conjugate prior distribution of the categorical distribution. Typically, a symmetric Dirichlet distribution is chosen, reflecting ignorance about which states are inherently more likely than others. The single parameter of this distribution (termed the concentration parameter) controls the relative density or sparseness of the resulting transition matrix. A choice of 1 yields a uniform distribution. Values greater than 1 produce a dense matrix, in which the transition probabilities between pairs of states are likely to be nearly equal. Values less than 1 result in a sparse matrix in which, for each given source state, only a small number of destination states have non-negligible transition probabilities. It is also possible to use a two-level prior Dirichlet distribution, in which one Dirichlet distribution (the upper distribution) governs the parameters of another Dirichlet distribution (the lower distribution), which in turn governs the transition probabilities. The upper distribution governs the overall distribution of states, determining how likely each state is to occur; its concentration parameter determines the density or sparseness of states. Such a two-level prior distribution, where both concentration parameters are set to produce sparse distributions, might be useful for example in unsupervised part-of-speech tagging, where some parts of speech occur much more commonly than others; learning algorithms that assume a uniform prior distribution generally perform poorly on this task. The parameters of models of this sort, with non-uniform prior distributions, can be learned using Gibbs

sampling or extended versions of the expectation-maximization algorithm. An extension of the previously described hidden Markov models with Dirichlet priors uses a Dirichlet process in place of a Dirichlet distribution. This type of model allows for an unknown and potentially infinite number of states. It is common to use a two-level Dirichlet process, similar to the previously described model with two levels of Dirichlet distributions. Such a model is called a hierarchical Dirichlet process hidden Markov model, or HDP-HMM for short. It was originally described under the name "Infinite Hidden Markov Model" and was further formalized in "Hierarchical Dirichlet Processes".

>> Section 3

A hidden Markov model (HMM) is a Markov model in which the observations are dependent on a latent (or hidden) Markov process (referred to as X $\{\displaystyle X\}$). An HMM requires that there be an observable process Y $\{\displaystyle Y\}$ whose outcomes depend on the outcomes of X $\{\displaystyle X\}$ in a known way. Since X $\{\displaystyle X\}$ cannot be observed directly, the goal is to learn about state of X $\{\displaystyle X\}$ by observing Y $\{\displaystyle Y\}$. By definition of being a Markov model, an HMM has an additional requirement that the outcome of Y $\{\displaystyle Y\}$ at time $t = t_0$ $\{\displaystyle t=t_0\}$ must be "influenced" exclusively by the outcome of X $\{\displaystyle X\}$ at $t = t_0$ $\{\displaystyle t=t_0\}$ and that the outcomes of X $\{\displaystyle X\}$ and Y $\{\displaystyle Y\}$ at $t < t_0$ $\{\displaystyle t<t_0\}$ must be conditionally independent of Y $\{\displaystyle Y\}$ at $t = t_0$ $\{\displaystyle t=t_0\}$ given X $\{\displaystyle X\}$ at time $t = t_0$ $\{\displaystyle t=t_0\}$. Estimation of the parameters in an HMM can be performed using maximum likelihood estimation. For linear chain HMMs, the Baum–Welch algorithm can be used to estimate parameters. Hidden Markov models are known for their applications to thermodynamics, statistical mechanics, physics, chemistry, economics, finance, signal processing, information theory, pattern recognition—such as speech recognition, handwriting recognition, gesture recognition, part-of-speech tagging, musical score following, partial discharges and bioinformatics.

>> Section 4

Learning The parameter learning task in HMMs is to find, given an output sequence or a set of such sequences, the best set of state transition and emission probabilities. The task is usually to derive the maximum likelihood estimate of the parameters of the HMM given the set of output sequences. No tractable algorithm is known for solving this problem exactly, but a local maximum likelihood can be derived efficiently using the Baum–Welch algorithm or the

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Baldi–Chauvin algorithm. The Baum–Welch algorithm is a special case of the expectation-maximization algorithm. If the HMMs are used for time series prediction, more sophisticated Bayesian inference methods, like Markov chain Monte Carlo (MCMC) sampling are proven to be favorable over finding a single maximum likelihood model both in terms of accuracy and stability. Since MCMC imposes significant computational burden, in cases where computational scalability is also of interest, one may alternatively resort to variational approximations to Bayesian inference, e.g. Indeed, approximate variational inference offers computational efficiency comparable to expectation-maximization, while yielding an accuracy profile only slightly inferior to exact MCMC-type Bayesian inference.

>> Section 5

General state spaces In the hidden Markov models considered above, the state space of the hidden variables is discrete, while the observations themselves can either be discrete (typically generated from a categorical distribution) or continuous (typically from a Gaussian distribution). Hidden Markov models can also be generalized to allow continuous state spaces. Examples of such models are those where the Markov process over hidden variables is a linear dynamical system, with a linear relationship among related variables and where all hidden and observed variables follow a Gaussian distribution. In simple cases, such as the linear dynamical system just mentioned, exact inference is tractable (in this case, using the Kalman filter); however, in general, exact inference in HMMs with continuous latent variables is infeasible, and approximate methods must be used, such as the extended Kalman filter or the particle filter. Nowadays, inference in hidden Markov models is performed in nonparametric settings, where the dependency structure enables identifiability of the model and the learnability limits are still under exploration.